



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science  
Faculty of Computing

## Project Task 3

( *Choosing The Appropriate LAN Devices* )

|                     |                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programme</b>    | Bachelor of Computer Science ( <i>Data Engineering</i> )                                                                                         |
| <b>Subject Code</b> | SECR1213                                                                                                                                         |
| <b>Subject Name</b> | Network Communications                                                                                                                           |
| <b>Session-Sem</b>  | 2024/2025-1                                                                                                                                      |
| <b>Prepared by</b>  | 1) ANIS SAFIYYA BIN JANAI<br>2) NURUL ASYIKIN BINTI KHAIRUL ANUAR<br>3) DAMIYA AINA BINTI BASIR ABD SHAMMAD<br>4) AIN NURNABILA BINTI MOHD AZHAR |
| <b>Section</b>      | 02                                                                                                                                               |
| <b>Group</b>        | CircuitCrew (CC)                                                                                                                                 |
| <b>Lecturer</b>     | DR. ISMAIL FAUZI BIN ISNIN                                                                                                                       |
| <b>Date</b>         | 20 / 11 / 2024                                                                                                                                   |

## Table of Content

|                                      |           |
|--------------------------------------|-----------|
| <b>List of Devices.....</b>          | <b>3</b>  |
| <b>List of Suitable Devices.....</b> | <b>4</b>  |
| <b>Group Reflection.....</b>         | <b>7</b>  |
| <b>Minutes meeting #1.....</b>       | <b>8</b>  |
| <b>Minutes Meeting #2.....</b>       | <b>9</b>  |
| <b>References.....</b>               | <b>10</b> |

## LIST OF DEVICES

|                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Damiya Aina binti Basir Abd Shammad</p> <ol style="list-style-type: none"> <li>1. Aruba CX 6300 Series</li> <li>2. Ruckus R750 (Wi-Fi 6)</li> <li>3. Huawei CloudEngine S12700 Series</li> <li>4. CloudEngine S16700 Series Switches</li> <li>5. Fortinet FortiGate 80F</li> </ol>                  | <p>Ain Nurnabila binti Mohd Azhar</p> <ol style="list-style-type: none"> <li>1. Cisco Catalyst 1000</li> <li>2. TP-Link JetStream</li> <li>3. Aruba Instant On AP22</li> <li>4. Juniper switches EX2300 / EX3400 Series</li> </ol>                                                                 |
| <p>Nurul Asyikin binti Khairul Anuar</p> <ol style="list-style-type: none"> <li>1. TP-Link Archer BE550</li> <li>2. Netgear Nighthawk RS700S</li> <li>3. L-Com TRD795BLK-15M</li> <li>4. TERA GRAND CAT7-7000-14W</li> <li>5. ARRIS SURFboard SB8200</li> <li>6. NETGEAR Cable Modem CM2000</li> </ol> | <p>Anis Safiyya binti Janai</p> <ol style="list-style-type: none"> <li>1. Cisco ISR 1100 Series</li> <li>2. Cisco Aironet 2800/3800</li> <li>3. Dell PowerEdge R340</li> <li>4. Asus RT-AX88U</li> <li>5. Prolink UTP Cat6 Full Copper Cable</li> <li>6. DbillionDa Cat8 Ethernet Cable</li> </ol> |

## LIST OF SUITABLE DEVICES

### 1. Aruba CX 6300 Series ( Core Switches )



Price range : RM35,167.00

Quantity : 2 units

This modern, flexible, and intelligent stackable switch series is ideal for enterprise network access, aggregation, core, and data center top of rack (ToR) deployments.

Key benefits :

- **Simpler network operations** - Save time and eliminate VLAN sprawl by reducing the configuration needed for SSIDs, ACLs, subnets, and wired ports
- **Enhanced security and visibility** - Ensure users and devices only communicate with destinations consistent with their mission via centralized policy definition and role-based access
- **Cloud-based management and automation** - Leverage intent-based, easy-to-use workflows for policy definition and network configuration
- **Performance and efficiency** - Eliminate IT overhead with automatically updated and continuously enforced policy

### 2. TP-Link Archer BE550 ( Access Switches )



Price range : RM 869.00

Quantity : 10 units

This router is a tri-band Wi-Fi 7 router with speeds up to 9.2Gbps, that is ideal for small and medium settings. This router consists of 5 ports which is 1 LAN and 4 WAN, with the speed 2.5Gbps each.

Key benefits:

- **Wi-Fi 7 Compatibility** - Supports the most recent Wi-Fi 7 standard, which provides incredibly fast speeds and lower latency to meet the demands of current networking.
- **High Data Throughput** - Equipped with multi-gigabit Ethernet connections (up to 2.5G or more), guaranteeing uninterrupted connectivity and data transfer.
- **Easy Management** - Allows for easy network management and setting using TP-Link's Tether App and web user interface.
- **Flexible Connectivity** - Provides smooth interaction with current networks and backward compatibility with older Wi-Fi standards.
- **Sustainability** - Energy-efficient design reduces power usage while maintaining performance.

### 3. Ruckus R750 (Wi-Fi 6) (Wireless Access Point )



Price : RM1710.00

Quantity : 15 units

The R750 makes it easy to deliver reliable, secure, high-performance connectivity in transit hubs, auditoriums, conference centers and other high traffic indoor spaces.

#### Key Benefits:

- **Connect more devices simultaneously** - Improve device performance, by enabling more simultaneous device connections with built-in 8 spatial streams, MU-MIMO and OFDMA technology.
- **High density performance** - Provides exceptional end-user experience within large meeting halls, general enterprise spaces, and large classrooms with the RUCKUS Ultra-High-Density Technology Suite.
- **Enhanced Security** - Enhanced security The latest Wi-Fi security standard with WPA3 and receive enhanced protection from man-in-the-middle attacks in the most secure way
- **More Than Wi-Fi** - Support services beyond Wi-Fi with RUCKUS IoT Suite, Cloudpath security and onboarding software, SPoT WiFi locationing engine, and SCI network analytic

### 4. Fortinet FortiGate 80F ( Firewall and Security Network )



Price : RM6136.00

Quantity : 1 unit

The FortiGate and FortiWiFi 80F series integrate firewalling, SD-WAN, and security in one appliance, making them excellent for constructing secure networks across distributed enterprise sites and transforming WAN architecture at any scale.

#### Key Benefits:

- Simplified operations with centralized management for networking and security, automated workflows, deep analytics, and self-healing
- Inclusive SD-WAN and wireless controller in every FortiGate appliance at no extra cost
- Rich portfolio for any business budget and need

## 5. L-Com TRD795BLK-15M



Price range : RM 807.00

Quantity : 500 pairs

The L-Com TRD795BLK-15M is a high-performance Category 7 (Cat7) Ethernet cable intended for demanding network environments. It has great signal integrity and can transport data at speeds of up to 10 Gbps over a 15 metre distance. The cable has Shielded Foiled Twisted Pair (S/FTP) shielding to reduce electromagnetic interference (EMI) and crosstalk, making it ideal for areas with high electrical noise.

### Key Benefits:

- **High-speed Performance** - Perfect for applications requiring a lot of bandwidth, it supports data transmission rates of up to 10 Gbps.
- **Reliable Signal Integrity** - S/FTP shielding creates connections that are reliable and free from interference by lowering EMI and crosstalk.
- **Future Proof** - Compatibility with the newest networking standards and devices is guaranteed by Cat7 compliance.
- **Seamless Performance** - Ready-to-use RJ45 connections for simple integration with routers, switches, and other network equipment.

## GROUP REFLECTION

1. Are you surprised by the prices? How were you surprised?

Yes, we were surprised, especially by how much enterprise-grade networking equipment costs. At first glance, devices like Aruba switches and Fortinet firewalls seem really expensive compared to home or small business routers. These devices aren't just about providing a simple connection, they're built for reliability, security, and handling large volumes of traffic so we think that it is reasonable to be that expensive. For example, a Fortinet firewall doesn't just protect the network, it actively monitors threats in real time. That kind of functionality costs money.

2. Have you ever considered cost as a factor for choosing networking devices?

Definitely. Cost is always a major consideration, especially with a budget like RM1.5M. While that sounds like a lot, networking gear adds up fast when we're planning for a large setup. Choosing the right balance of performance and affordability is key.

For example, we discussed Ruckus wireless access points, which are excellent but on the pricier side. If we went for a more budget-friendly brand like Ubiquiti, we'd save money, but we might sacrifice some performance in areas with heavy user traffic. Similarly, Huawei offers great functionality at a lower cost compared to Cisco, making it a solid option if we are looking for value without completely compromising on quality.

Ultimately, we're aiming to spend smartly, invest in areas where performance and security are critical, and save where we can without jeopardizing reliability.

3. What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers

The differences usually boil down to performance, cost, and ecosystem. Here's what we noticed when comparing Cisco and Huawei:

- **Performance and Features:** Cisco routers are often seen as the gold standard. They're highly reliable, packed with advanced features (like enhanced security and VPN support), and perform well in high-demand environments. Huawei, on the other hand, offers competitive performance but at a lower cost. They might not have all the bells and whistles Cisco has, but they're still solid.
- **Cost:** Huawei is generally more budget-friendly. This makes it a great choice for setups like ours, where we need quality but also need to stay within budget. Cisco, while expensive, offers excellent long-term support and stability, which can justify the cost in critical setups.

## Meeting Minutes #1

|               |                                   |                                                                                                    |                    |
|---------------|-----------------------------------|----------------------------------------------------------------------------------------------------|--------------------|
| Date/Time     |                                   | 17 Nov 2024 / 5.00 pm                                                                              |                    |
| Location      |                                   | Foyer Dewan Resak                                                                                  |                    |
| Agenda        |                                   | Task Distribution for Task 3                                                                       |                    |
| Meeting Host  |                                   | Damiya Aina binti Basir Abd Shammad                                                                |                    |
| Attendance    |                                   |                                                                                                    |                    |
| Name          |                                   | Time                                                                                               | Reason For Absence |
| Anis Safiyya  |                                   | 5.00 pm                                                                                            | -                  |
| Nurul Asyikin |                                   | 5.00 pm                                                                                            | -                  |
| Damiya Aina   |                                   | 5.00 pm                                                                                            | -                  |
| Ain Nurnabila |                                   | 5.00 pm                                                                                            | -                  |
| Minutes       |                                   |                                                                                                    |                    |
| No.           | Item Discussed                    | Details                                                                                            | Person-In-Charge   |
| 1.            | Find devices for case study       | All members were assigned to find at least 2 devices based on the floor plan on previous task      | All members        |
| 2.            | Research about the devices        | All members need to gather some information about the devices for reflection                       | All members        |
| 3.            | Task Distribution                 | Damiya told Ain, Anis and Asyikin to find the devices each                                         | Damiya Aina        |
| 3.            | Methods for information gathering | - Ain , Damiya , Anis and Asyikin were gather and list the devices before choose the most suitable | All members        |



## Meeting Minutes #2

|               |                                                   |                                                                                          |                    |
|---------------|---------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|
| Date/Time     |                                                   | 19 Nov 2024 / 9.30 pm                                                                    |                    |
| Location      |                                                   | Whatsapp Chat                                                                            |                    |
| Agenda        |                                                   | Progress Discussion                                                                      |                    |
| Meeting Host  |                                                   | Damiya Aina binti Basir Abd Shammad                                                      |                    |
| Attendance    |                                                   |                                                                                          |                    |
| Name          |                                                   | Time                                                                                     | Reason For Absence |
| Anis Safiyya  |                                                   | 9.30 pm                                                                                  | -                  |
| Nurul Asyikin |                                                   | 9.30 pm                                                                                  | -                  |
| Damiya Aina   |                                                   | 9.30 pm                                                                                  | -                  |
| Ain Nurnabila |                                                   | 9.30 pm                                                                                  | -                  |
| Minutes       |                                                   |                                                                                          |                    |
| No.           | Item Discussed                                    | Details                                                                                  | Person-In-Charge   |
| 1.            | Sharing session on current progress               | All of the members shared the current progress                                           | All members        |
| 2.            | Discuss the most suitable devices for the project | - All members share ideas and opinions of what devices that will be used for the project | All members        |
| 3.            | Final edit in report                              | - Each member completing the additional information in report                            | All members        |
| 4.            | Meeting ended                                     | At 11.00                                                                                 | Damiya Aina        |

## REFERENCES

1. Ruckus Network  
*R750 | RUCKUS R750 Indoor Access Point | RUCKUS Networks.* (n.d.). RUCKUS Networks. <https://www.ruckusnetworks.com/products/wireless-access-points/r750/>
2. Aruba Network  
Hewlett Packard Enterprise. (n.d.). *HPE OfficeConnect 1920S Switch Series* [Document A00058593ENW]. Retrieved November 19, 2024, from <https://www.hpe.com/psnow/doc/a00058593enw>
3. Wireless LAN  
*Top rated Wireless LAN vendors.* (n.d.). PeerSpot. <https://www.peerspot.com/categories/wireless-lan?rr=t>
4. Huawei Routers  
*CloudEngine S16700 Series Switches | Huawei Enterprise.* (n.d.). Huawei Enterprise. <https://e.huawei.com/en/products/switches/campus-switches/s16700>
5. Fortinet Networks  
*FortiGate FortiWiFi 80F Series Data Sheet | Fortinet.* (n.d.). Fortinet. <https://www.fortinet.com/resources/data-sheets/fortigate-fortiwifi-80f-series>
6. TP-Link Archer BE550  
TP-Link. (n.d.). *BE9300 Tri-Band Wi-Fi 7 router.* <https://www.tp-link.com/my/home-networking/wifi-router/archer-be550/>
7. L-Com TRD795BLK-15M  
L-Com. (n.d.). *Category 7 10Gig Ethernet cable assembly, S/FTP braid with individually shielded pairs, RJ45 male plug, 26AWG stranded PVC, black, 15.0m.* Retrieved November 17, 2024, from <https://www.l-com.com/category-7-10gig-ethernet-cable-assembly-s-ftp-braid-with-individual-ly-shielded-pairs-rj45-male-plug-26awg-stranded-pvc-black-150m>
8. NETGEAR Cable Modem CM2000  
Parrish, K., & Parrish, K. (2024, October 25). *NETGEAR Nighthawk RS700S Review.* HighSpeedInternet.com. [https://www.highspeedinternet.com/resources/netgear-nighthawk-rs700s-review#:~:text=Ports :%20\\*%204x%20Gigabit%20Ethernet%20LAN%20ports,\(back\)%20\\*%201x%20USB%20DA %203.2%20port%20\(back\)](https://www.highspeedinternet.com/resources/netgear-nighthawk-rs700s-review#:~:text=Ports,%20*%204x%20Gigabit%20Ethernet%20LAN%20ports,(back)%20*%201x%20USB%20DA%203.2%20port%20(back))